Deep Learning–Based Diagnosis of Polycystic Ovary Syndrome Using Clinical Data and Ultrasound Imaging of Sri Lankan Patients

Research proposal

Submitted in partial fulfilment of the requirement for the degree of Bachelor of Science Honours in Computer Science

By:
Anjalee Kaushalya
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Department of Physical Science
Faculty of Applied Science
University of Vavuniya
Sri Lanka

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1 Declaration

I, **Lokusooriyage Anjalee Kaushalya** (2020/ASP/67), do hereby declare that the work reported in this project thesis was carried out independently by me under the supervision of **Dr R.Nagulan, Senior Lecture, Department of Physical science, Faculty of applied science, University of Vavuniya** and **Dr. Peshala Dangalla, Consultant Obstetrician and Gynecologist, Colombo North Teaching Hospital, Ragama**. This thesis describes the results of my own research, except where due reference has been made in the text. No part of this research project has been submitted earlier or concurrently for the same or any other degree.

Date: ______________ Signature: ________________________

Certified by:

1. Supervisor: Dr. R.Nagulan Date: ______________
Signature: ________________________

2. Supervisor: Dr. Peshala Dangalla Date: 15.01.2026
Signature: ________________________

Approved by Coordinator/ CSH4116
Name: Ms. S.Subaramya Date: ______________
Signature: ________________________

Approved by Head/ Department of Physical Science
Name: Mr. N. Edwin Linosh Date: ______________
Signature: ________________________

2 Introduction

2.1 Background

Polycystic Ovary Syndrome (PCOS) is a medical condition that causes hormonal disorders in women in their childbearing years. PCOS is a common gynecological endocrine disorder that affects approximately 5-10 % of women aged 18 to 44 years [12].PCOS is a growing women's health concern in Sri Lanka, with both community and clinic-based studies highlighting its prevalence and complications. A community study in the Gampaha District (2005–2006) involving 3,030 women aged 15–39 reported a 5.9% prevalence of confirmed PCOS aligning with international figures [10]. Similarly, a genetic study identified a 6.3% prevalence among premarital Sri Lankan women, showing that PCOS manifests early and is linked to metabolic risks such as obesity, insulin resistance, and dyslipidemia [5]. Clinical studies from Colombo [15] show that women with PCOS commonly present with menstrual irregularities, hyperandrogenism, and subfertility, alongside high rates of obesity and insulin resistance. Case–control studies further reveal increased risks of diabetes, hypertension, and metabolic syndrome compared to healthy women [20].

These findings indicate that PCOS in Sri Lanka not only affects women of reproductive age at rates consistent with global prevalence (6%), but also carries a substantial metabolic and reproductive health burden. This highlights the urgent need for early diagnosis, culturally appropriate interventions, and monitoring to prevent long-term complications such as diabetes and cardiovascular disease among affected women. .

Early diagnosis of PCOS is vital as it enables timely management of reproductive issues such as menstrual irregularities and subfertility, while also preventing long-term complications including insulin resistance, type 2 diabetes, cardiovascular disease, and endometrial cancer. It further supports psychological wellbeing by addressing symptoms like acne, hirsutism, and weight gain, thereby improving overall quality of life.

PCOS is diagnosed through a combination of clinical evaluation, hormonal tests, and imaging studies . Clinically, physicians assess menstrual irregularities and signs of hyperandrogenism, while laboratory tests measure hormone levels and metabolic parameters. Ultrasound imaging is particularly important because it allows direct visualization of the ovaries, identifying the characteristic polycystic morphology, such as multiple small follicles or increased ovarian volume, which is a key diagnostic feature. Ultrasound is non-invasive, widely accessible, and provides objective evidence that complements clinical and laboratory findings, making it central to confirming a PCOS diagnosis.

Traditional diagnosis relies on manual interpretation of hormonal tests, ultrasound images, and clinical features, which can be time-consuming, subjective, and prone to variability [1]. Machine learning (ML) and deep learning (DL) offer significant advantages in the diagnosis of PCOS by automating and enhancing the accuracy of detecting complex patterns in clinical, biochemical, and imaging data [1]. Deep learning, in particular, is well-suited for analyzing complex time-series health data and imaging data, such as ovarian ultrasound scans, allowing for automated and objective assessment.

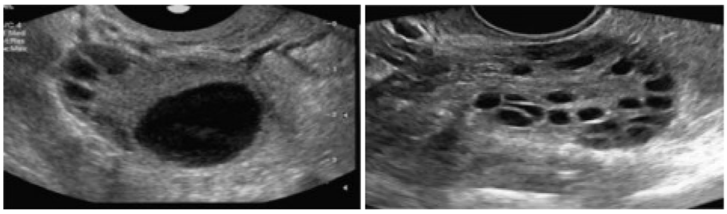


Figure 1: **Ultrasound images of normal and PCOS affected ovary [17].**

Building on this, our study proposes a data-driven deep learning framework for PCOS diagnosis using a combination of health records and ultrasound imaging. Data collection will include hospital datasets from the General Hospital in the Gampaha District (with ethical approval), clinical datasets (menstrual cycles), and Ultrasound images. For model design, we will employ a multimodal deep learning approach to predict upcoming menstrual cycles and learn normal patterns over time, and a Convolutional Neural Network (CNN) to extract features from ovarian ultrasound scans. The proposed deep learning framework can also be extended to categorize PCOS risk levels (Low, Moderate, High). Integrating these models into a combined architecture will enable a comprehensive and automated framework for accurate and early PCOS diagnosis.

2.2 Problem Definition

Despite its prevalence, PCOS often remains underdiagnosed or is diagnosed late due to the subjective nature of manual evaluation. The current diagnosis depends on clinical examination, hormonal testing, and ultrasound imaging, which require expert interpretation and can vary between practitioners.

Traditional approaches rely heavily on manual interpretation of clinical symptoms and ultrasound features, making them time-consuming and prone to variability [1]. Furthermore, healthcare settings in Sri Lanka face challenges such as limited access to specialists and fragmented health data, which hinder consistent diagnosis.

As a result, many women suffer from delayed diagnosis, untreated reproductive complications, and increased risk of long-term conditions such as type 2 diabetes, cardiovascular disease, and infertility. Hence, there is an urgent need for an efficient, data-driven diagnostic framework that can support clinicians in identifying PCOS accurately and early.

2.3 Research Objectives

The primary objective of this research is to design and develop a deep learning–based diagnostic framework for Polycystic Ovary Syndrome (PCOS) using clinical and ultrasound imaging data from Sri Lankan patients. The study aims to collect and preprocess menstrual health records and ovarian ultrasound images to build a multimodal model that integrates both temporal and visual information. A multimodel feature for menstrual cycle pattern analysis and a Convolutional Neural Network (CNN) for ultrasound image classification will be implemented. Through this approach, the research seeks to improve the accuracy and efficiency of PCOS diagnosis, detect abnormalities in menstrual patterns, and categorize patients into different risk levels (low, moderate, and high). Ultimately, the project intends to support clinicians in making timely, data-driven decisions for better reproductive health outcomes.

2.4 Motivation

PCOS is a common hormonal disorder that causes infertility, metabolic issues, and long-term health risks. Current diagnosis methods mainly manual ultrasound interpretation and fragmented health records are often subjective and inconsistent. In Sri Lanka, early detection is further challenged by limited datasets and healthcare resources. Deep learning offers a powerful solution by integrating health records and ultrasound imaging for accurate and automated diagnosis, while also enabling risk-level assessment to guide clinical decisions.

3 related works

Table 1: Previous Studies of PCOS Using Machine Learning and Deep Learning Approaches (Part 1)

Reference	**Dataset**	**ML/DL Algorithm**	**Limitations**	**Highest Performance**
[18]	Ultrasoundimages .com and Sonoworld.com	CNN	Dataset not clearly described	85%
[6]	Custom dataset (40+14 images)	CNN	Small dataset; low testing performance	100%
[9]	Ultrasound images (Kaggle)	CNN	Limited dataset; lacks diversity	98%
[4]	Kaggle images	DCNN	Feature selection only; no performance reported	–

Table 2: Previous Studies of PCOS Using Machine Learning and Deep Learning Approaches (Part 2)

Reference	Dataset	ML/DL Algorithm	Limitations	Highest Performance
[3]	Custom ultrasound dataset	VGG16, VGG19, InceptionV3, MobileNet, DenseNet121/201	Limited computational resources; noisy images excluded	82%
[19]	Ultrasound images	ANN + Adaptive K-means clustering	Limited details provided	75%
[11]	68 custom ultrasound images	NB, ANN	Small dataset size	100%
[8]	Numeric dataset (42 variables)	SVM, CART, NB, RF, LR	Small dataset; preprocessing not described	98%
[16]	Kaggle numeric data (541 samples)	RF, KNN, SVM, NN, NB	Unbalanced dataset	93%
[14]	Kaggle data (43 attributes of 541 samples)	SVM	Unbalanced dataset	91%
[7]	20 patient ultrasound scans	SVM	Very small dataset	95%
[2]	Kaggle data (43 attributes of 541 samples)	SVM, KNN, NB, Ensemble (SVM+KNN)	Unbalanced dataset	98%
[13]	Kaggle numeric data (43 attributes of 541 samples)	DT, LDA, LR, KNN, LSVM, QSVM, CSVM, FGSV, MGSV, CGSVM	Unbalanced dataset	100%

4 Research Gap

Although PCOS is common in Sri Lanka, its diagnosis still relies on subjective manual evaluation of symptoms, hormone tests, and ultrasound images. Existing deep learning studies are mostly based on foreign datasets, use small or unbalanced samples, and often analyze clinical or imaging data separately. Very few studies use time-series menstrual data or focus on Sri Lankan women's unique clinical patterns. Therefore, there is a clear research gap in developing an integrated deep learning model that combines menstrual, clinical, and ultrasound data for accurate and early PCOS diagnosis in the Sri Lankan healthcare context.

5 Materials and Methods

This study follows a quantitative, data-driven experimental design that incorporates clinical records, menstrual health data, and ultrasound images of Sri Lankan women. A multimodal deep learning approach is used. The final diagnostic output is generated by integrating both models.

Procedures

- Data Collection :
  - Hospital Data : Clinical records and ovarian ultrasound images will be collected from routine clinic visits at Colombo North Teaching Hospital, Ragama, and District General Hospitals in Gampaha and Colombo, with appropriate ethical approval. The data may be existing or generated as part of this study.
  - Menstrual/Clinical Data : Information on menstrual cycle patterns, hormonal levels, BMI, and other health indicators will be collected.
  - Ultrasound Images :
    - Normal ovaries : From women without PCOS.
    - PCOS-affected ovaries : From diagnosed patients.
  - Participants aged 18 years and above may enroll directly. For participants under 18 years, written consent from a parent or legal guardian will be obtained before participation.
- Data Management and Post-Research Handling:
  - All collected data will be stored securely on password-protected and encrypted devices, with access restricted to the research team only. Personally identifiable information will be anonymized at the earliest stage of data processing.

- After completion of the research, anonymized datasets will be securely retained under the custody of the Department, in accordance with approved ethical guidelines. Access to the data will be controlled and permitted only for ethically approved academic or verification purposes. Any personal or identifiable data will be permanently destroyed using secure deletion methods for electronic files and shredding for physical records. Anonymized datasets will be archived only if ethically permitted; otherwise, they will also be permanently deleted.

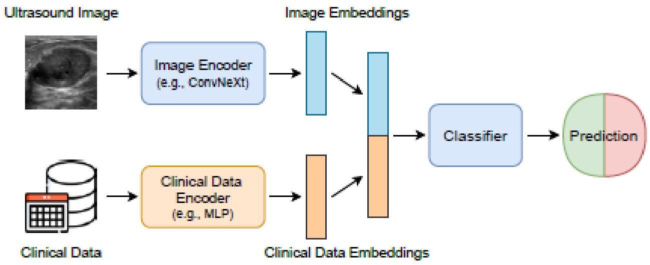
- Data Preprocessing :
  - Ultrasound images will be prospectively collected during routine clinical examinations, minimizing quality variations between older and newly acquired scans.
  - All images will be obtained using VOG (transvaginal ultrasound) following standard clinical protocols during the study period.
  - To maintain consistency, ultrasound images will be:
    - Acquired using the same or clinically equivalent ultrasound equipment.
    - Captured according to routine standardized clinical scanning procedures.
    - Collected within the same clinical workflow, ensuring comparable resolution and imaging parameters.
  - Missing Data : Handled using simple imputation (e.g., forward fill, interpolation).
  - normalize data, handle missing values, and format it as time-series for deep learning models.
  - Obtain ovarian ultrasound images labeled with diagnostic outcomes (e.g., PCOS, normal).

- Feature Extraction :
  - Image Branch: Use a convolutional neural network (CNN) backbone, such as ResNet-18, ConvNeXt, or VGG-16

    Input: Preprocessed ultrasound images (e.g., resized to 224x224, normalized), Output: Flattened feature embeddings (e.g., a 1D vector of 512-2048 dimensions)
  - Numerical Data Branch: Employ a multilayer perceptron (MLP) with 2-3 fully connected layers (e.g., input size matching the number of features $\rightarrow$ 16 $\rightarrow$ 8 unit

    Input: Normalized numerical features (e.g., BMI, hormonal levels, TSH, FBS, HDL, LDL; handle categorical aspects via one-hot encoding if needed).

    Output: Compact embeddings (e.g., 8-16 dimensions)



- Multimodel Feature Fusion :

  Use intermediate fusion at the feature level Concatenation or attention-based multimodal fusion.

- Model Training :
  - Train the model on labeled datasets (normal vs abnormal).
  - Use output probabilities to categorize data into risk groups (low, moderate, high risk).

- Prediction :
  - Accurate and early PCOS diagnosis for Sri Lankan patients.

6 Expected Outcome

1. Early detection of PCOS using AI accurately
2. Combines clinical data and ultrasound imaging for better insights.
3. Identifies at-risk individuals early for timely clinical intervention.
4. Enables personalized treatment and faster diagnosis.

7 Ethical Approval

This study will be conducted in accordance with established ethical standards. Ethical approval will be obtained from the Ethics Review Committees of Colombo North Teaching Hospital, District General Hospital Gampaha, and other Colombo . All clinical records and ultrasound images will be anonymized before analysis, and personal identifiers will be removed to ensure confidentiality. Data will be stored securely and used only for research purposes. Informed consent will be obtained for any data collected directly from participants. These measures ensure that the study protects patient privacy and follows responsible research practices.

8 Budget

No.	Item Description	Details / Justification	Estimated Cost (LKR)	
1	Travel to Ragama (Colombo Teaching Hospital)	5 round trips from Gampaha to ragama (bus/train + local transport) for data coordination and discussions with Consultant Obstetrician & Gynecologistt	LKR	2,000.00
2	External storage device	Secure encrypted external hard drive (1 TB) for dataset storage	LKR	18,000.00
3	Printing and stationery	Printing of data request letters, proposal copies, and related documents.	LKR	5,000.00
4	Research Assistant support	Part-time voluntary assistance for dataset annotation and administrative tasks.	LKR	2,500.00
	Total		**LKR**	**27,500.00**

Figure 2: **All expenses are self-funded and personally borne by the researcher. No external funding or university resources are required. A part-time research assistant will provide voluntary support, with any token honorarium covered personally**

9 Gant Chart

Work/Time (Months)	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	May-26	Jun-26
Title selection	■							
Proposal		■						
Literature Review		■	■	■				
Resource gathering			■	■				
Tools and Techniques			■	■				
Implementation				■	■	■		
Writing			■	■	■	■	■	
Publication / Conferences							■	
Presentation and Submission								■

References

[1] S. Ahmed et al. "A review on the detection techniques of polycystic ovary syndrome using machine learning". In: *IEEE Access* 11 (2023), pp. 86522–86543.

[2] M. Alagarsamy et al. "Detection of polycystic syndrome in ovary using machine learning algorithm". In: *International Journal of Intelligent Systems and Applications in Engineering* 11.1 (2023), pp. 246–253.

[3] A. Alamoudi et al. "A deep learning fusion approach to diagnose the polycystic ovary syndrome (PCOS)". In: *Applied Computational Intelligence and Soft Computing* 2023 (2023), pp. 1–15.

[4] S. Bhosale, L. Joshi, and A. Shivsharan. *PCOS (Polycystic Ovarian Syndrome) Detection Using Deep Learning*. Online. Accessed: May 16, 2022. 2022.

[5] U. Branavan, S. Wijesundera, and V. Chandrasekharan. "Potential genetic polymorphisms predicting polycystic ovary syndrome (PCOS) in Sri Lankan women: Comparison with different ethnicity". In: (2021). Unpublished manuscript or institutional report.

[6] B. Cahyono, M. S. Mubarok, and U. N. Wisesty. "An implementation of convolutional neural network on PCO classification based on ultrasound image". In: *Proceedings of the 5th International Conference on Information and Communication Technology (ICoIC)*. 2017, pp. 1–4.

[7] S. S. Deshpande and A. Wakankar. "Automated detection of polycystic ovarian syndrome using follicle recognition". In: *Proceedings of the IEEE International Conference on Advanced Communication, Control and Computing Technologies*. 2014, pp. 1341–1346.

[8] M. M. Hassan and T. Mirza. "Comparative analysis of machine learning algorithms in diagnosis of polycystic ovarian syndrome". In: *International Journal of Computer Applications* 175.17 (2020), pp. 42–53.

[9] A. K. M. S. Hosain, M. H. K. Mehedi, and I. E. Kabir. "PCONet: A convolutional neural network architecture to detect polycystic ovary syndrome (PCOS) from ovarian ultrasound images". In: *arXiv preprint arXiv:2210.00407* (2022).

[10] V. Kumarapeli et al. "A simple screening approach for assessing community prevalence and phenotype of polycystic ovary syndrome in a semi-urban population in Sri Lanka". In: *American Journal of Epidemiology* 168.3 (2008), pp. 321–328. DOI: 10.1093/aje/kwn137.

[11] K. Maheswari et al. "Modelling of F3I based feature selection approach for PCOS classification and prediction". In: *Journal of Ambient Intelligence and Humanized Computing* 12.1 (2021), pp. 1349–1362.

[12] U. A. Ndefo, A. Eaton, and M. R. Green. “Polycystic ovary syndrome: a review of treatment options with a focus on pharmacological approaches”. In: *P T* 38.6 (2013), pp. 336–355.

[13] E. Nsugbe. “An artificial intelligence-based decision support system for early diagnosis of polycystic ovaries syndrome”. In: *Healthcare Analytics* 3 (2023), p. 100164.

[14] B. Purnama et al. “A classification of polycystic ovary syndrome based on follicle detection of ultrasound images”. In: *Proceedings of the 3rd International Conference on Information and Communication Technology (ICoICT)*. 2015, pp. 396–401.

[15] I. Ranathunga et al. “Evaluation of socio-demographic and clinical characteristics of PCOS patients attending a tertiary care institute in Colombo”. In: *BMC Endocrine Disorders* 22.1 (2022), p. 289.

[16] C. R. N. Satish, X. Chew, and K. W. Khaw. “Polycystic ovarian syndrome (PCOS) classification and feature selection by machine learning techniques”. In: *Applied Mathematics and Computational Intelligence (AMCI)* 9 (2020), pp. 65–74.

[17] K. Shanmugavadivel et al. “Optimized polycystic ovarian disease prognosis and classification using AI-based computational approaches on multi-modality data”. In: *BMC Medical Informatics and Decision Making* 24.1 (2024), p. 281.

[18] M. Sumathi et al. “Study and detection of PCOS related diseases using CNN”. In: *IOP Conference Series: Materials Science and Engineering*. Vol. 1070. 2021, p. 012062.

[19] L. Thara and T. M. Divya. “Detection and prediction system for polycystic ovary syndrome using structural normalized square similarity detection approach”. In: *Nveo-Natural Volatiles and Essential Oils Journal* 8.6 (2021), pp. 2834–2842.

[20] K. Weerasinghe. “Cardiovascular risk factors among women with polycystic ovary syndrome in an urban Sri Lankan population”. In: *Endocrine Abstracts*. Vol. 81. Bioscientifica, 2022.